Exercise 10 – Distributed Systems

Create a Distributed System

The goal of this exercise is to create and run a distributed system composed of the following components:

1. **Nginx proxy**
2. **MariaDB database**
3. **Backend application**

Each component should run in its own dedicated Docker container. You can use **Docker Compose** to define and manage the system.

Nginx Proxy

- Use any version of the official nginx image.
- The Nginx server should be accessible in your browser at: **https://localhost/**
- The root path (/) should serve the provided HTML/CSS/JS files.
- All other paths should automatically redirect to the backend.
- Communication must be over **HTTPS** (a self-signed certificate is acceptable).
- You will need to:
  - Create a **custom Nginx configuration**.
  - Use this configuration inside the Docker container.
  - Ensure that **only the proxy is exposed** to the outside world.

MariaDB Database

- Use the official mariadb image
- The database will be used to store and retrieve data from the backend.
- You must create an **initialization SQL script** that sets up:
  - A database.
  - A table to store user information with the following fields:
    * `id`: Auto-incrementing integer
    * `username`: Text
    * `email`: Text
    * `phone`: Text
    * `profile_picture_path`: Text

Backend Application

You are free to use any language or framework for the backend. It should implement the following endpoints:

1. `POST /person/`: Should handle the content of the html form and save it to the database
2. `GET /persons/`: Should retrive all created members and serialize them in a JSON file